Thermo Fisher SCIENTIFIC

# SAFETY DATA SHEET

## Section 1 - Identification

**Product Identifier**

| | |
|---|---|
| **Product Name** | **Salmonella H c Agglutinating Sera** |
| **Recommended Use** | In vitro diagnostic. |
| **Uses advised against** | No Information available |

| | |
|---|---|
| **Product Code** | **R30160401** |
| **Address** | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| **Emergency Tel.** | **CHEMTREC®**<br>**09 980 6780 or +64 9 980 6780** |
| **Telephone / Fax Numbers** | Tel: 09 980 6700<br>Fax: 09 980 6788 |
| **E-mail address** | ANZinfo@thermofisher.com |

## Section 2 - Hazard(s) Identification

**Classification under Work Safe New Zealand**

**Not classified as hazardous according to criteria of EPA New Zealand**

**GHS Classification**

**Physical hazards**
Based on available data, the classification criteria are not met

**Health hazards**
Based on available data, the classification criteria are not met

**Environmental hazards**
Based on available data, the classification criteria are not met

**Label Elements** None required

**Other hazards which do not result in classification**
This product does not contain any known or suspected endocrine disruptors

# Section 3 - Composition and Information on Ingredients

| Component | CAS No | Weight % |
|---|---|---|
| Sodium hydroxide | 1310-73-2 | <0.5 |
| Sodium azide | 26628-22-8 | 0.1 |

# Section 4 - First Aid Measures

**Description of first aid measures**

**New Zealand Emergency Tel.** CHEMTREC®
09 980 6780 or +64 9 980 6780

**Inhalation** Remove to fresh air. Get medical attention.

**Eye Contact** Rinse thoroughly with plenty of water, also under the eyelids. Get medical attention.

**Skin Contact** Get medical attention. Wash with plenty of soap and water.

**Ingestion** Do NOT induce vomiting. Get medical attention.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**First Aid Facilities** Eyewash, safety shower and washroom.

**Most important symptoms and effects** No information available.

**Notes to Physician** Treat symptomatically.

# Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**
No information available.

**Specific Hazards Arising from the Chemical**
Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**
None under normal use conditions.

**Special protective equipment and precautions for fire fighters**
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

# Section 6 - Accidental Release Measures

**Personal Precautions, Protective Equipment and Emergency Procedures**

**Emergency procedures**
Use personal protective equipment as required. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing.

**Environmental Precautions**
Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. See Section 12 for additional Ecological

Information.

**Methods for Containment and Clean Up**
Soak up with inert absorbent material. Clean contaminated surface thoroughly.

**Precautions to prevent secondary hazards**
Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**
Refer to protective measures listed in Sections 8 and 13.

# Section 7 - Handling and Storage

## Precautions for Safe Handling

**Advice on safe handling**
Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only in well-ventilated areas.

**Hygiene Measures**
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## Conditions for Safe Storage, Including any Incompatibilities

**Storage Conditions**
Keep container tightly closed. Keep at temperatures between 2°C and 8 °C.

**Incompatible Materials**
Strong oxidizing agents. Acids.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

# Section 8 - Exposure Controls and Personal Protection

## Control parameters

**Exposure limits**
**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor
**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]
Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia
**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.
**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

| Component | New Zealand WEL | Australia | ACGIH TLV | The United Kingdom |
|---|---|---|---|---|
| Sodium hydroxide | Ceiling: 2 mg/m³ | 2 mg/m³ TWA | Ceiling: 2 mg/m³ | 2 mg/m³ STEL |
| Sodium azide | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m³ | CL 0.11 ppm (0.3 mg/m³) | Ceiling: 0.29 mg/m³<br>Ceiling: 0.11 ppm | Skin<br>TWA 0.1 mg/m³<br>STEL 0.3 mg/m³ |

**Biological limit values**
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Appropriate engineering controls

**Engineering Measures**
Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation

or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments |
|---|---|---|---|---|
| Disposable gloves. | See manufacturers recommendations | - | AS/NZS 2161 | (minimum requirement) |

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Repiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of repiratory protective devices (or AUS/NZ equivalent)
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains.

# Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

**Physical State** Liquid

| | | |
|---|---|---|
| **Appearance** | Amber | |
| **Odor** | No information available | |
| **Odor Threshold** | No data available | |
| **pH** | 6.6 - 6.8 | |
| **Melting Point/Range** | Not applicable | |
| **Softening Point** | No data available | |
| **Boiling Point/Range** | Not applicable | |
| **Flammability (liquid)** | No data available | |
| **Flammability (solid,gas)** | No information available | |
| **Explosion Limits** | No data available | |
| **Flash Point** | Not applicable | **Method -** No information available |
| **Autoignition Temperature** | Not applicable | |
| **Decomposition Temperature** | No data available | |
| **Viscosity** | No data available | |
| **Water Solubility** | No information available | |
| **Solubility in other solvents** | No information available | |
| **Partition Coefficient (n-octanol/water)** | | |
| **Vapor Pressure** | No data available | |
| **Density / Specific Gravity** | No data available | |
| **Bulk Density** | No data available | |
| **Vapor Density** | No data available | (Air = 1.0) |
| **Particle characteristics** | Not applicable (liquid) | |

**Other information**

# Section 10 - Stability and Reactivity

**Reactivity** None known, based on information available

**Stability** Stable under recommended storage conditions.

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

**Conditions to Avoid** Extremes of temperature and direct sunlight.

**Incompatible Materials** Strong oxidizing agents, Acids.

**Hazardous Decomposition Products** None under normal use conditions.

# Section 11 - Toxicological Information

**Acute Effects**

**Information on likely routes of exposure**

**Product Information** Product does not present an acute toxicity hazard based on known information

**Inhalation** Not an expected route of exposure.
**Eyes** Not an expected route of exposure.
**Skin** No known effect based on information supplied.
**Ingestion** No known effect based on information supplied.

**Numerical measures of toxicity**

**(a) acute toxicity;**
**Oral** Based on ATE data, the classification criteria are not met
**Dermal** Based on ATE data, the classification criteria are not met
**Inhalation** No data available

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|---|---|---|---|
| Sodium hydroxide | LD50 = 325 mg/kg ( Rat ) | LD50 = 1350 mg/kg ( Rabbit ) | |
| Sodium azide | LD50 = 27 mg/kg ( Rat ) | - | LC50 0.054 - 0.52 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**
**Respiratory** No data available
**Skin** No data available

**Sensitization** None known

**(e) germ cell mutagenicity;** No data available

None known

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available
**Reproductive Effects** None known
**Developmental Effects** None known

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects,both acute and delayed**
No information available.

# Section 12 - Ecological Information

**Ecotoxicity**

**Aquatic ecotoxicity** .

| Component | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|---|---|---|---|---|
| Sodium hydroxide | LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss) | - | - | - |
| Sodium azide | LC50: = 0.7 mg/L, 96h (Lepomis macrochirus) LC50: = 0.8 mg/L, 96h (Oncorhynchus mykiss) LC50: = 5.46 mg/L, 96h flow-through (Pimephales promelas) | | | |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** No information available

**Bioaccumulative Potential** No information available

**Mobility** No information available.

**Other adverse effects**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

| | |
|---|---|
| **Persistent Organic Pollutant** | This product does not contain any known or suspected substance |
| **Ozone Depletion Potential** | This product does not contain any known or suspected substance |

# Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products**
Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**
Empty containers should be taken to an approved waste handling site for recycling or disposal.

**Other Information**
Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations .

# Section 14 - Transport Information

| Component | Hazchem Code |
|---|---|
| Sodium hydroxide<br>1310-73-2 ( <0.5 ) | 2W<br>2R |
| Sodium azide<br>26628-22-8 ( 0.1 ) | 2XE |

**NZS 5433:2020**
Not regulated

**IATA**
Not regulated

**IMDG/IMO**
Not regulated

**Environmental hazards**
No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**
Not applicable, packaged goods

**Special Precautions**
No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information**
None known

# Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National Regulations**
There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**
Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most

explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**
Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---|---|---|---|
| Sodium hydroxide | - | Use restricted. See item 75.<br>(see link for restriction details) | - |

*https://echa.europa.eu/substances-restricted-under-reach*

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | NZIoC | AICS | EINECS | ELINCS | NLP | KECL | IECSC | TCSI |
|---|---|---|---|---|---|---|---|---|---|
| Sodium hydroxide | 1310-73-2 | X | X | 215-185-5 | - | - | KE-31487 | X | X |
| Sodium azide | 26628-22-8 | X | X | 247-852-1 | - | - | KE-31357 | X | X |

| Component | CAS No | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|---|---|---|---|---|---|---|---|---|
| Sodium hydroxide | 1310-73-2 | X | ACTIVE | X | - | X | X | X |
| Sodium azide | 26628-22-8 | X | ACTIVE | X | - | X | X | X |

**Legend:** X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

# Section 16 - Other Information

## This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List
**IECSC** - Chinese Inventory of Existing Chemical Substances
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances
**TWA** - Time Weighted Average
**AICS** - Australian Inventory of Chemical Substances
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
**ENCS** - Japanese Existing and New Chemical Substances
**KECL** - Korean Existing and Evaluated Chemical Substances
**CAS** - Chemical Abstracts Service
**ACGIH** - American Conference of Governmental Industrial Hygienists

**IARC** - International Agency for Research on Cancer
**NZS 5433:2020** - Transport of Dangerous Goods on Land
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association
**MARPOL** - International Convention for the Prevention of Pollution from Ships
**LD50** - Lethal Dose 50%
**EC50** - Effective Concentration 50%
**WEL** - Workplace Exposure Limit
**DNEL** - Derived No Effect Level
**POW** - Partition coefficient Octanol:Water
**vPvB** - very Persistent, very Bioaccumulative
**VOC** - (Volatile Organic Compound)
**PNEC** - Predicted No Effect Concentration
**OECD** - Organisation for Economic Co-operation and Development
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code
**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail
**LC50** - Lethal Concentration 50%
**ATE** - Acute Toxicity Estimate
**RPE** - Respiratory Protective Equipment
**NOEC** - No Observed Effect Concentration
**BCF** - Bioconcentration factor
**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**
HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS
EPA Guide to classifying hazardous substances in New Zealand
EPA - Assigning a product to an existing HSNO approval guide

**Training Advice**
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

| | |
|---|---|
| **Revision Date** | 30-Jun-2023 |
| **Revision Summary** | Not applicable |

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

# End of Safety Data Sheet